

Recall $r \times c$ matrix A defines a map $f = f_A$
from $\mathbb{R}^c$ to $\mathbb{R}^r$ by eqⁿ $f(x) = Ax$

To solve $Ax=b$ i.e. $f(x)=b$, row reduce $A|b$ to
its reduced row echelon form RREF. Two cases

- Either system is inconsistent $\Leftrightarrow$ pivot in last col of $\text{RREF}(A|b)$
- OR $\left\{ \begin{array}{l} \text{the "dim" of solution space} = \# \text{ of non-pivot columns in} \\ \text{RREF}(A) = \# \text{ free variables} \end{array} \right.$
and then (general solution to $Ax=b$)
 $= (\text{a fixed sol}^n \text{ to } Ax=b) + (\text{gen sol}^n \text{ to } Ax=0)$

The following are equivalent TFAE / TFAE

- | | |
|---|---|
| ① f_A is injective (i.e. 1 to 1) | ①' f_A is surjective (i.e. onto) |
| ② $Ax=0$ has only the 0 sol ⁿ | ②' $Ax=b$ is consistent for each $b \in \mathbb{R}^r$ |
| ③ Every col of $\text{RREF}(A)$ has a pivot | ③' Every row in $\text{RREF}(A)$ has a pivot |
| ④ Col vectors of $A \dots$ | ④' Col vectors of $A \dots$ |
| ⑤ Row vectors of $A \dots$ | ⑤' Row vectors of $A \dots$ |
- will fill these later

Remarks

- | | |
|---|------------------|
| ① IF f_A is injective then $c \leq r$ | } Converse false |
| IF f_A is surjective then $r \leq c$ | |
| IF f_A is bijective then $r = c$ | |

② If $r=c$ i.e. if A is square matrix, then ③ $\Leftrightarrow$ ③'
 $\therefore$ 10 conditions (6 as of now) are equivalent!

In particular

• f inj $\Leftrightarrow f$ surjective and then f is bijective, so invertible
 $\Leftrightarrow \text{RREF}(A) = I \Leftrightarrow \underbrace{E_N \cdots E_2 E_1}_E A = I$ where E_i are elementary matrices
 $\Leftrightarrow EA = I$ for $E = \text{product of elementary matrices}$

How to find E ? Just do the same sequence of row operations (corresponding to $E_1, E_2, \dots, E_N$) to the identity matrix.

Defⁿ A matrix A is called invertible if there is a matrix B such that $AB = I$ $BA = I$

A priori all we know is $\left. \begin{array}{l} A \text{ } r \times c \\ B \text{ } c \times r \end{array} \right\} \begin{array}{l} AB = I_{r \times r} \\ BA = I_{c \times c} \end{array}$

But we'll soon see that this is possible only when $r=c$

Exercise ① If such B exists for A , then it is unique
(If B_1, B_2 are two such matrices, consider $(B_1 A) B_2 = B_1 (A B_2)$
So I can and do denote B by A^{-1}
② $(PQ)^{-1} = Q^{-1} P^{-1}$

Proof that A is invertible $\Rightarrow r=c$

Note for any matrices A, B (s.t. AB exists)

$$f_A \circ f_B(x) = f_A(Bx) = A(Bx) = (AB)x = f_{AB}(x)$$

$$\Rightarrow f_A \circ f_B = f_{AB}$$

So $AB = I \Rightarrow \boxed{f_A \circ f_B = f_I} = \text{identity map } \mathbb{R}^r \rightarrow \mathbb{R}^r$
 $BA = I \Rightarrow \boxed{f_B \circ f_A = f_I} = \text{identity map } \mathbb{R}^c \rightarrow \mathbb{R}^c$

$A \text{ } r \times c \quad B \text{ } c \times r$

$\mathbb{R}^c \xrightarrow{f_A} \mathbb{R}^r$
 $\xleftarrow{f_B}$

These eq^s say that f_A & f_B are both bijective

In particular f_A is injective & surjective $\Rightarrow r = c$ as claimed (end of proof)
 by table above

A & B are inverse matrices $\Leftrightarrow f_A$ and f_B are inverse functions

Upshot A invertible $\overset{?}{\Leftrightarrow} f_A$ bijective

$(\Leftrightarrow) (r = c \text{ \& } \text{REF}(A) = I)$

Exercise Prove $\Leftarrow$: If f_A is bijective, of course it has (a unique) inverse function. Why is this inverse $= f_E$ for some E ?

Exercise (i) Show that for square matrices A & B

$AB = I \Rightarrow BA = I$

(ii) Not true if A, B are not square. Find an example

Note For any set maps $f \circ g = \text{id} \Rightarrow f$ is surjective
 g is injective

Recall for tomorrow

- vector space
- subspace of a v.space: examples
 - subspaces of $\mathbb{R}^2, \mathbb{R}^3$
 - range and kernel of f_A for any matrix A

• Note: Range of $f_A = \{Ax \mid x \in \mathbb{R}^c\}$

IF $x = \begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix}$ $Ax = x_1 \begin{bmatrix} 1st \\ col of \\ A \end{bmatrix} + x_2 \begin{bmatrix} 2nd \\ col of A \end{bmatrix} + \dots + x_c \begin{bmatrix} last \\ col of A \end{bmatrix}$

Added after class

Exercise For a square matrix A , prove that TFAE.

- a) A is an invertible matrix
 - b) f_A is an invertible (i.e. bijective) function
 - c) $RREF(A) = \text{Identity matrix}$
 - d) $BA = I$ for some matrix B
 - e) $AC = I$ for some matrix C
 - f) $A = \text{product of elementary matrices}$
- and then $B=C$